

Supporting Information for

Combinatorial design of ionizable lipid nanoparticles for muscle-selective mRNA delivery with minimized off-target effects

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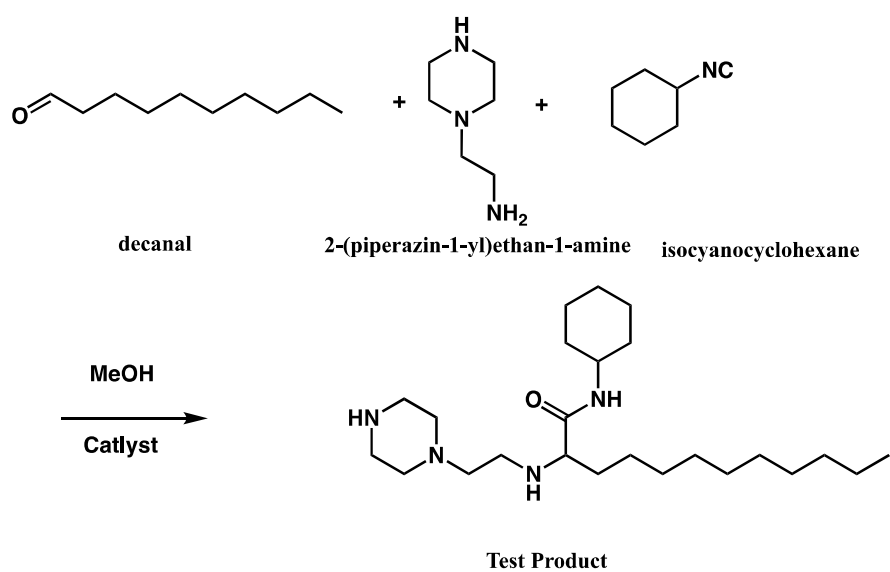
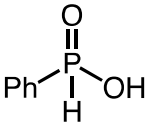


Fig S1. Scheme of preparation of test product.

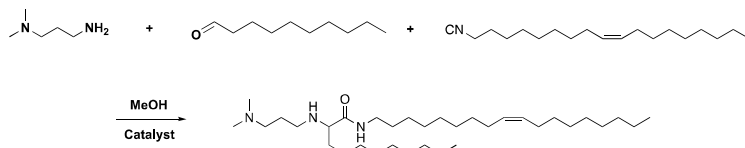
Table S1. Optimization of the reaction conditions of the test product.

Entry	Lewis acid	Condition	Solvent	Yield (%) ^a
1	N/A	rt/48	MeOH	10
2	TsOH (20%)	rt/24	MeOH	42
3	H ₃ PO ₄ (20%)	rt/24	MeOH	50
4	H ₃ BO ₄ (20%)	rt/24	MeOH	38
5	 (20%)	rt/24	MeOH	75

a: isolated yield.

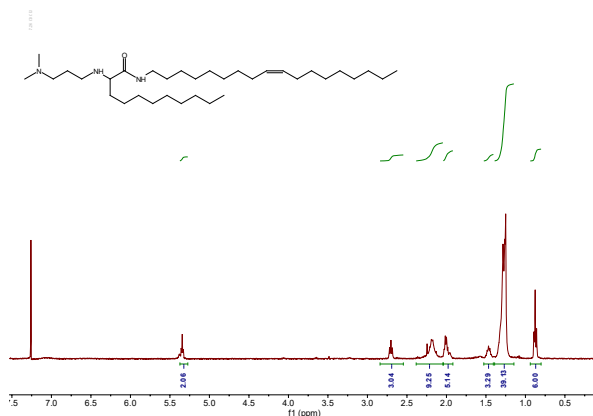
Fig. S2. Lipids synthesis in the panel (Page 4-6)

Lipid 1



Follow the synthesis method above. The residue was purified by silica gel chromatography (0.1% ammonia hydroxide with 0-10% MeOH in DCM) to give **lipid 1** (Yield: 75%, determined by combi flash of BUCHI purification system) as a colorless oil. ^1H NMR (400 MHz, CDCl_3) δ 5.37 – 5.27 (m, 2H), 2.70 (t, $J = 7.2$ Hz, 3H), 2.21 (d, $J = 24.8$ Hz, 9H), 2.04 – 1.92 (m, 5H), 1.46 (d, $J = 7.3$ Hz, 3H), 1.39 – 1.15 (m, 39H), 0.88 (t, $J = 6.7$ Hz, 6H).

^1H -NMR of lipid 1



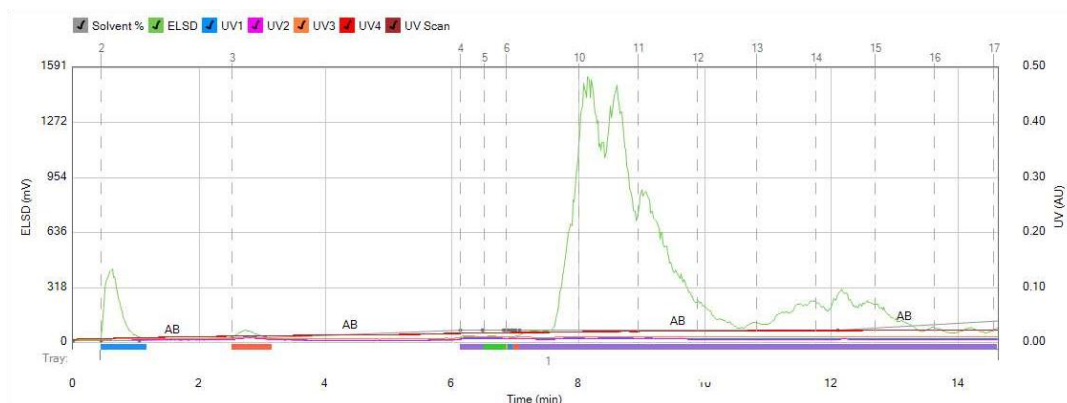
CombiFlash for lipid 1 purification.

Column: FP ECOFLEX Si 4g
Flow Rate: 15 mL/min
Equilibration: 2.4 min
Run Length: 14.7 min
Instrument Type: C-815
Mode: Flash
Sample type: Liquid

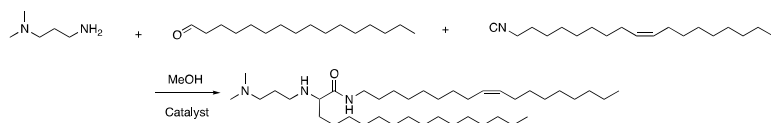
Solvent A: Dichloromethane
Solvent B: Methanol
Solvent C: Empty
Solvent D: Empty
Slope Detection: Off

UV Threshold: 0.05 AU
UV Sensitivity: Low
UV1 λ : 254 nm
UV2 λ : 265 nm
UV3 λ : 280 nm
UV4 λ : 320 nm
UV scan start λ : 254 nm (M)
UV scan end λ : 400 nm (M)

ELSD Threshold: 30 mV
ELSD Sensitivity: High
Collection: Collect Peaks
Per-Vial Volume: 14 mL
Non-Peak Volume: 14 mL

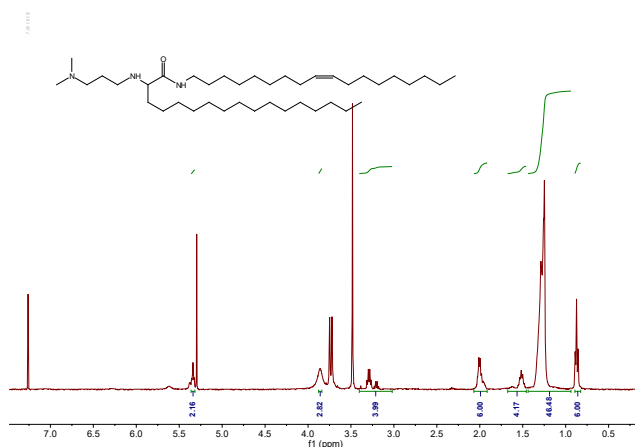


Lipid 2



Follow the synthesis method above. The residue was purified by silica gel chromatography (0.1% ammonia hydroxide with 0-10% MeOH in DCM) to give **lipid 2** (Yield: 90%, determined by combi flash of BUCHI purification system) as a colorless oil. ^1H NMR (400 MHz, CDCl_3) δ 5.36 – 5.32 (m, 2H), 3.86 (s, 3H), 3.25 (dq, $J = 34.4, 6.7$ Hz, 4H), 2.07 – 1.91 (m, 6H), 1.67 – 1.46 (m, 4H), 1.44 – 0.94 (m, 53H), 0.87 (t, $J = 6.7$ Hz, 6H).

^1H -NMR of lipid 2



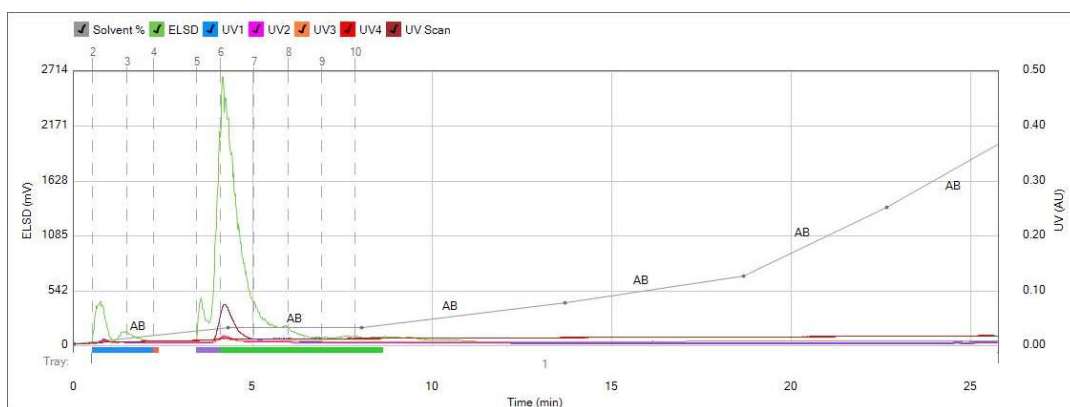
CombiFlash for lipid 2 purification.

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Equilibration: 2.4 min
Run Length: 25.8 min
Instrument Type: C-815
Mode: Flash
Sample type: Liquid

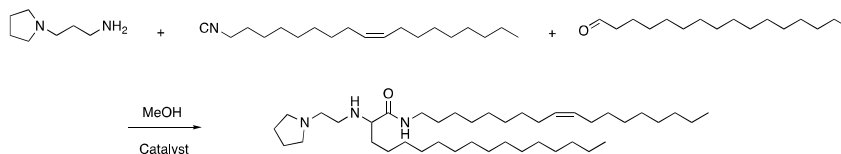
Solvent A: Dichloromethane
Solvent B: Methanol
Solvent C: Empty
Solvent D: Empty
Slope Detection: Off

UV Threshold: 0.05 AU
UV Sensitivity: Low
UV1 λ : 254 nm
UV2 λ : 265 nm
UV3 λ : 280 nm
UV4 λ : 320 nm
UV scan start λ : 254 nm (M)
UV scan end λ : 400 nm (M)

ELSD Threshold: 20 mV
ELSD Sensitivity: High
Collection: Collect Peaks
Per-Vial Volume: 14 mL
Non-Peak Volume: 14 mL

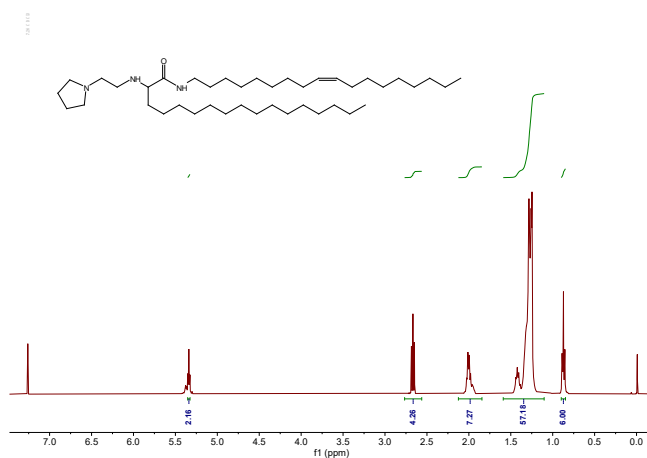


Lipid 3



Follow the synthesis method above. The residue was purified by silica gel chromatography (0.1% ammonia hydroxide with 0-10% MeOH in DCM) to give **lipid 3** (Yield: 85%, determined by combi flash of BUCHI purification system) as a colorless oil ^1H NMR (400 MHz, CDCl_3) δ 5.35 – 5.33 (m, 2H), 2.67 (t, $J = 7.0$ Hz, 4H), 2.00 (q, $J = 6.5$ Hz, 7H), 1.59 – 1.10 (m, 57H), 0.87 (t, $J = 6.7$ Hz, 6H).

^1H -NMR of lipid 3



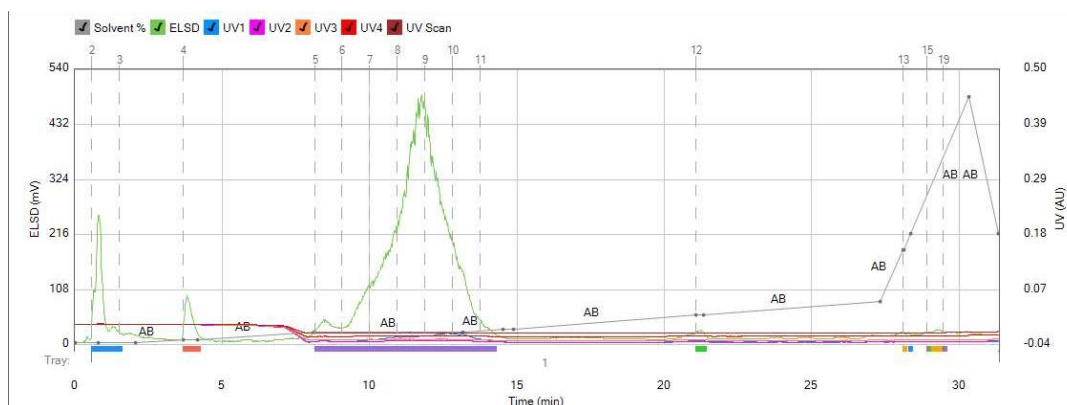
CombiFlash for lipid 3 purification.

Column: FP ECOFLEX Si 4g
 Flow Rate: 15 mL/min
 Equilibration: 2.4 min
 Run Length: 31.4 min
 Instrument Type: C-815
 Mode: Flash
 Sample type: Liquid

Solvent A: Dichloromethane
 Solvent B: Methanol
 Solvent C: Empty
 Solvent D: Empty
 Slope Detection: Off

UV Threshold: 0.05 AU
 UV Sensitivity: Low
 UV1 λ : 254 nm
 UV2 λ : 265 nm
 UV3 λ : 280 nm
 UV4 λ : 320 nm
 UV scan start λ : 254 nm (M)
 UV scan end λ : 400 nm (M)

ELSD Threshold: 20 mV
 ELSD Sensitivity: High
 Collection: Collect None
 Per-Vial Volume: 14 mL
 Non-Peak Volume: 14 mL



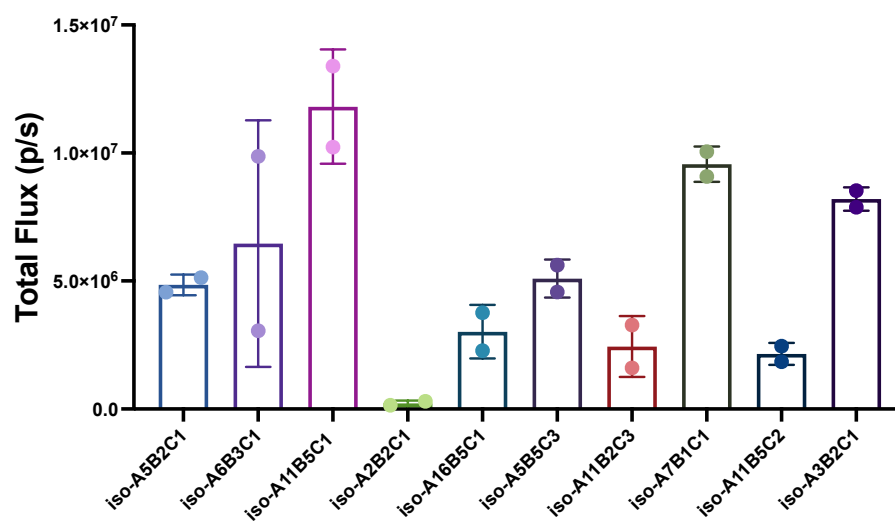
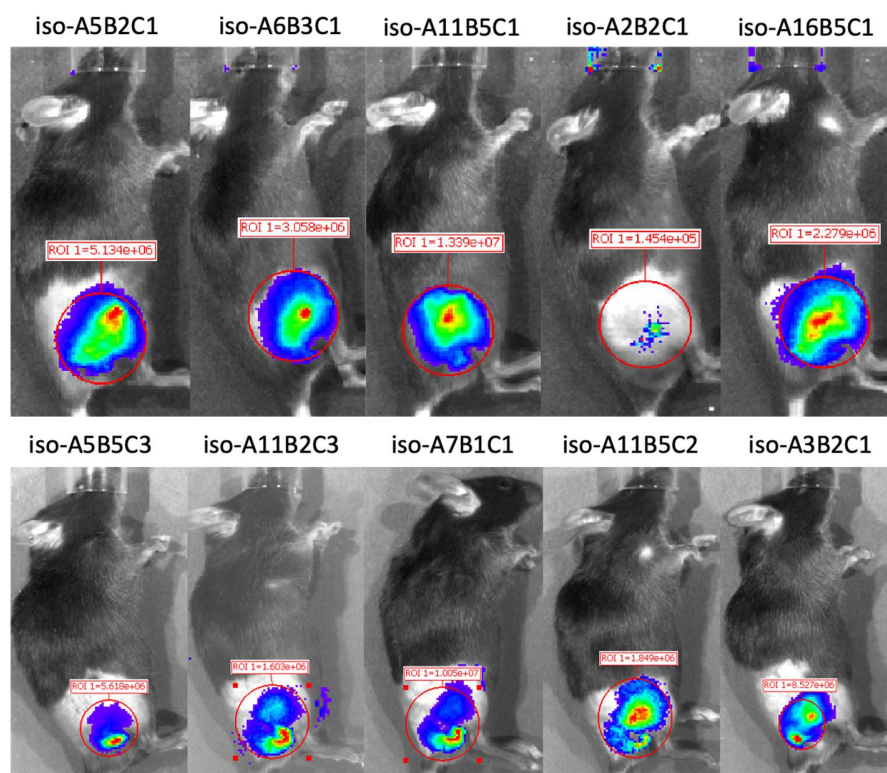


Fig. S3. *In vivo* screening of top 10 ionizable lipids from the lipid library.

Table S2. Formulations and characterization for optimizing iso-A11B5 LNP formulation.

LNP characteristics				LNP molar ratios (mol%)				Results	
ID	Ionizable lipid: mRNA weight ratio	Phospholipid type	Aqueous Buffer pH	Ionizable lipid content (mol%)	Phospholipid content (mol%)	Cholesterol content (mol%)	PEG content (mol%)	Encapsulation Efficiency (%)	Transfection potency (Log ₂ FC)
F-1	7.5	DOPE	3	60	15	24.5	0.5	79.82	13.84
F-2	12.5	DOPE	3	30	20	49.5	0.5	49.41	12.32
F-3	12.5	DOPE	6	30	15	54.5	0.5	77.57	13.84
F-4	7.5	DSPC	3	60	20	19.5	0.5	86.84	12.32
F-5	7.5	DSPC	6	45	10	43.5	1.5	87.50	15.61
F-6	7.5	DOPE	3	30	20	48.5	1.5	63.85	15.86
F-7	7.5	DOPE	6	30	10	59.5	0.5	60.58	17.43
F-8	10	DOPE	6	60	10	27.5	2.5	82.35	10.82
F-9	12.5	DSPC	6	30	20	49.5	0.5	82.08	14.49
F-10	10	DSPC	4.5	45	20	32.5	2.5	82.49	17.72
F-11	10	DOPE	4.5	45	15	37.5	2.5	81.37	11.81
F-12	12.5	DOPE	4.5	60	15	23.5	1.5	97.91	17.08
F-13	10	DSPC	3	30	15	53.5	1.5	89.30	17.10
F-14	7.5	DSPC	4.5	30	10	59.5	0.5	81.30	19.37
F-15	12.5	DOPE	3	45	10	44.5	0.5	54.59	13.43
F-16	7.5	DSPC	6	60	20	17.5	2.5	94.91	14.37
F-17	12.5	DSPC	6	60	20	17.5	2.5	98.17	9.22
F-18	12.5	DSPC	3	60	10	27.5	2.5	68.90	13.67

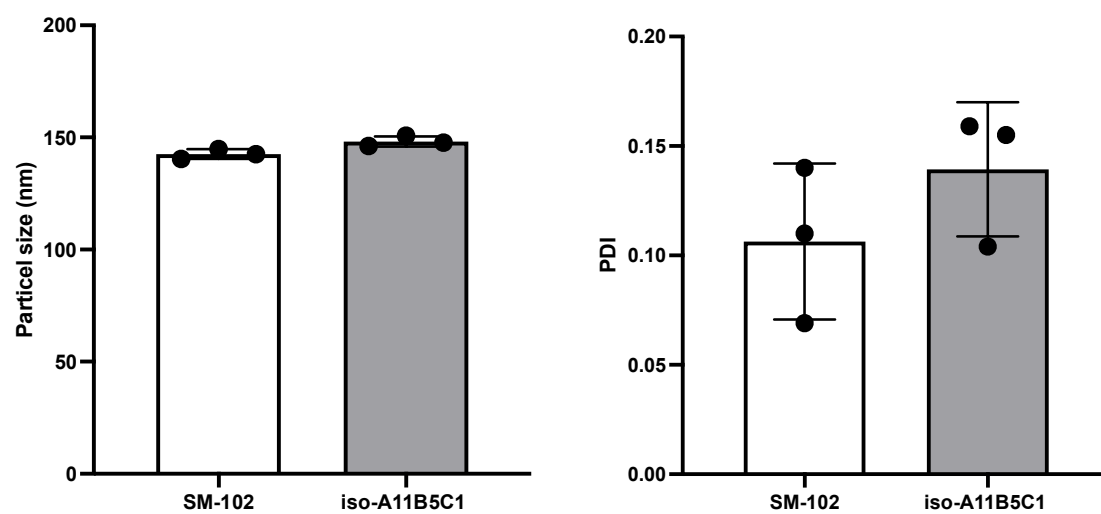


Fig. S4. LNP characterizations: particle size and PDI.

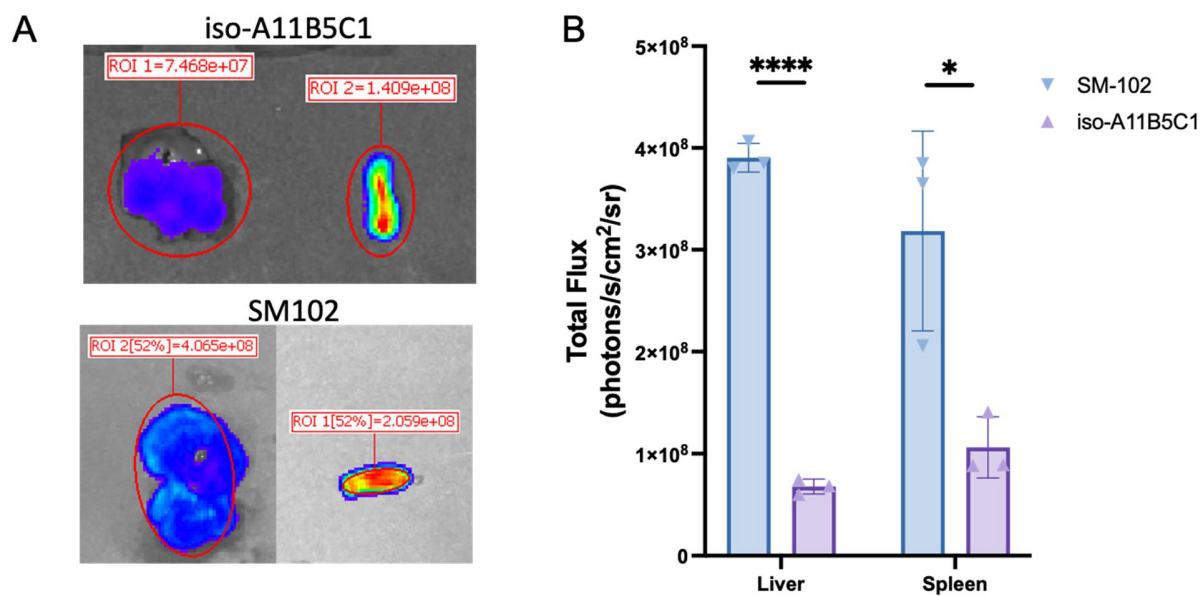


Fig. S5. Luminescence signal in the liver and spleen after intravenously injected with SM-102 or iso-A11B5C1 LNP/mLuc.

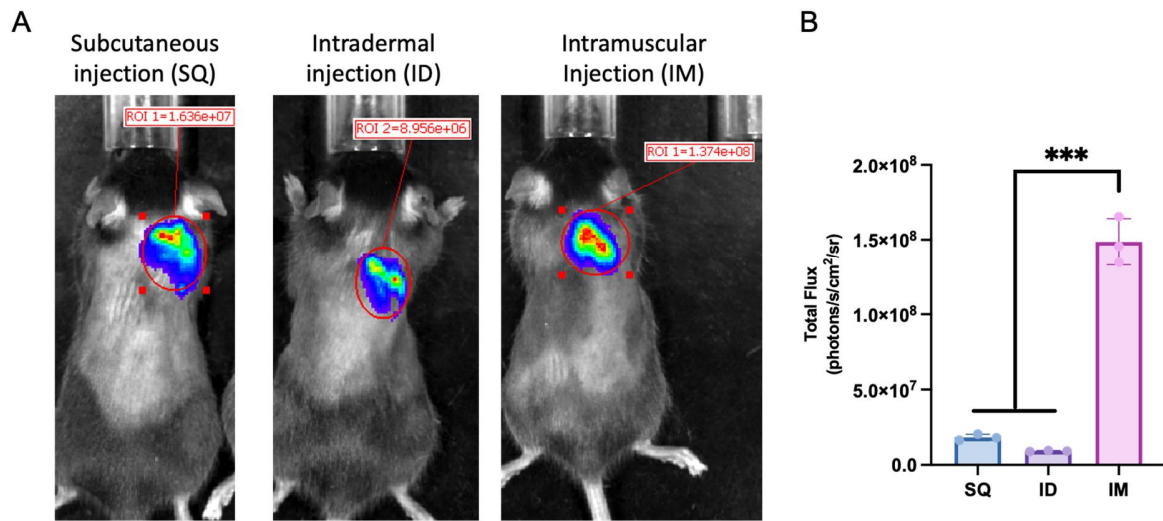


Fig. S6. Luminescence signal after injected with iso-A11B5C1 LNP/mLuc subcutaneously, intradermally, or intramuscularly.

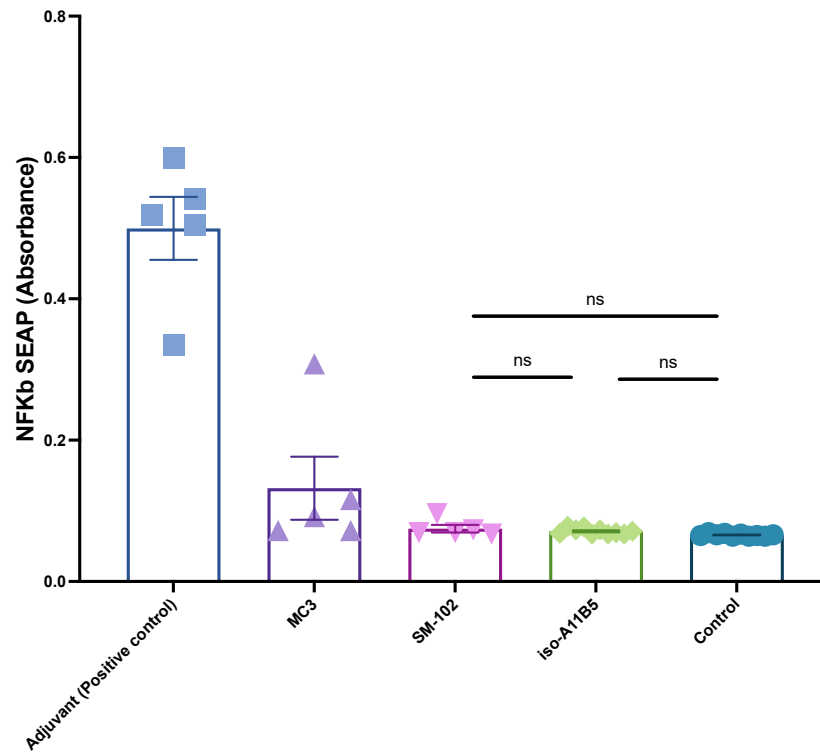


Fig. S7. Immunogenicity of iso-A11B5 LNP compared with MC3 LNP, SM102 LNP, measured by NFKb release level from THP-1 dual cells.

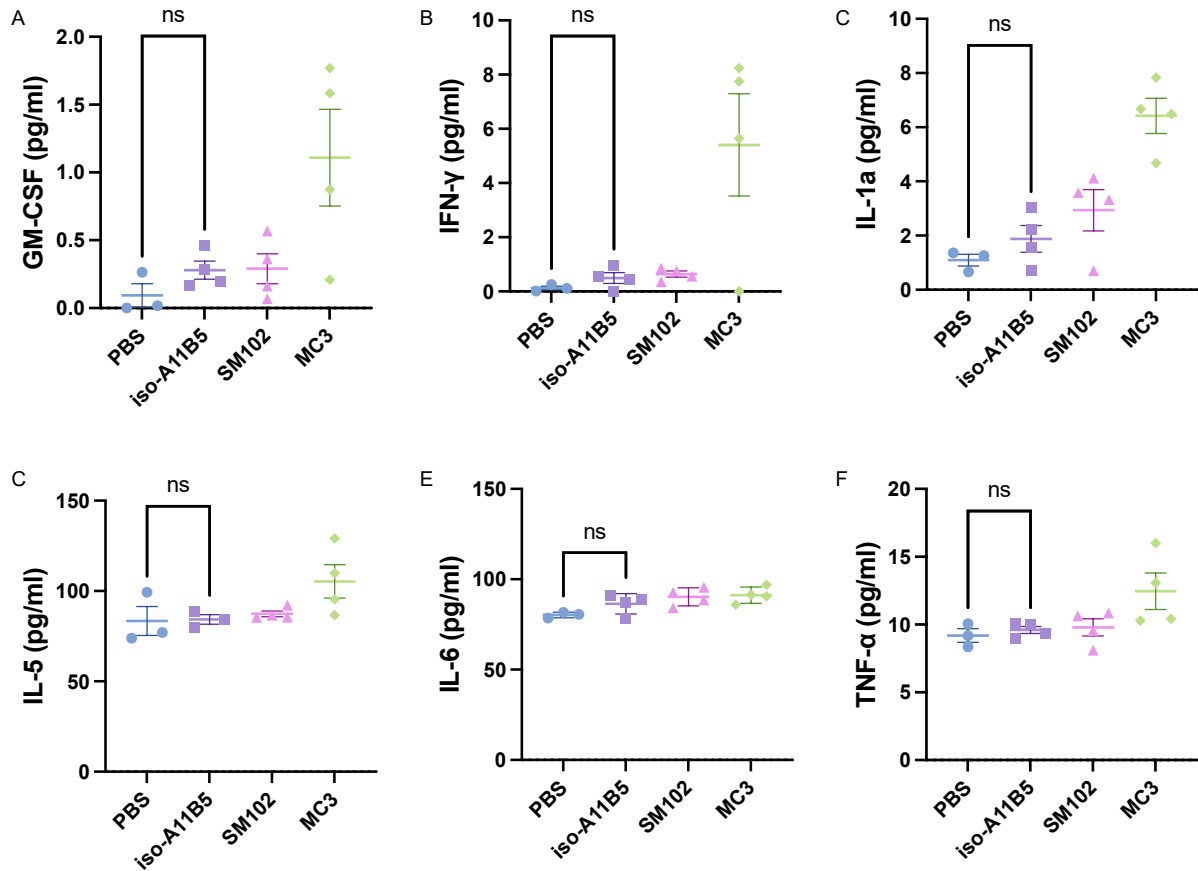


Fig. S8. Comparison of blood cytokines of mice 4h after I.M. injection of iso-A11B5C1, SM-102, and MC3 LNPs. 6-week-old female BALB/c mice were I.M. injected with different LNPs at the same dosage level (0.5 mg/kg, n=3 or 4). The control group was I.M. injected with PBS. Blood cytokines (A) GM-CSF, (B) IFN- γ , (C) IL-1a, (D) IL-5, (E) IL-6, and (F) TNF- α were compared between groups.

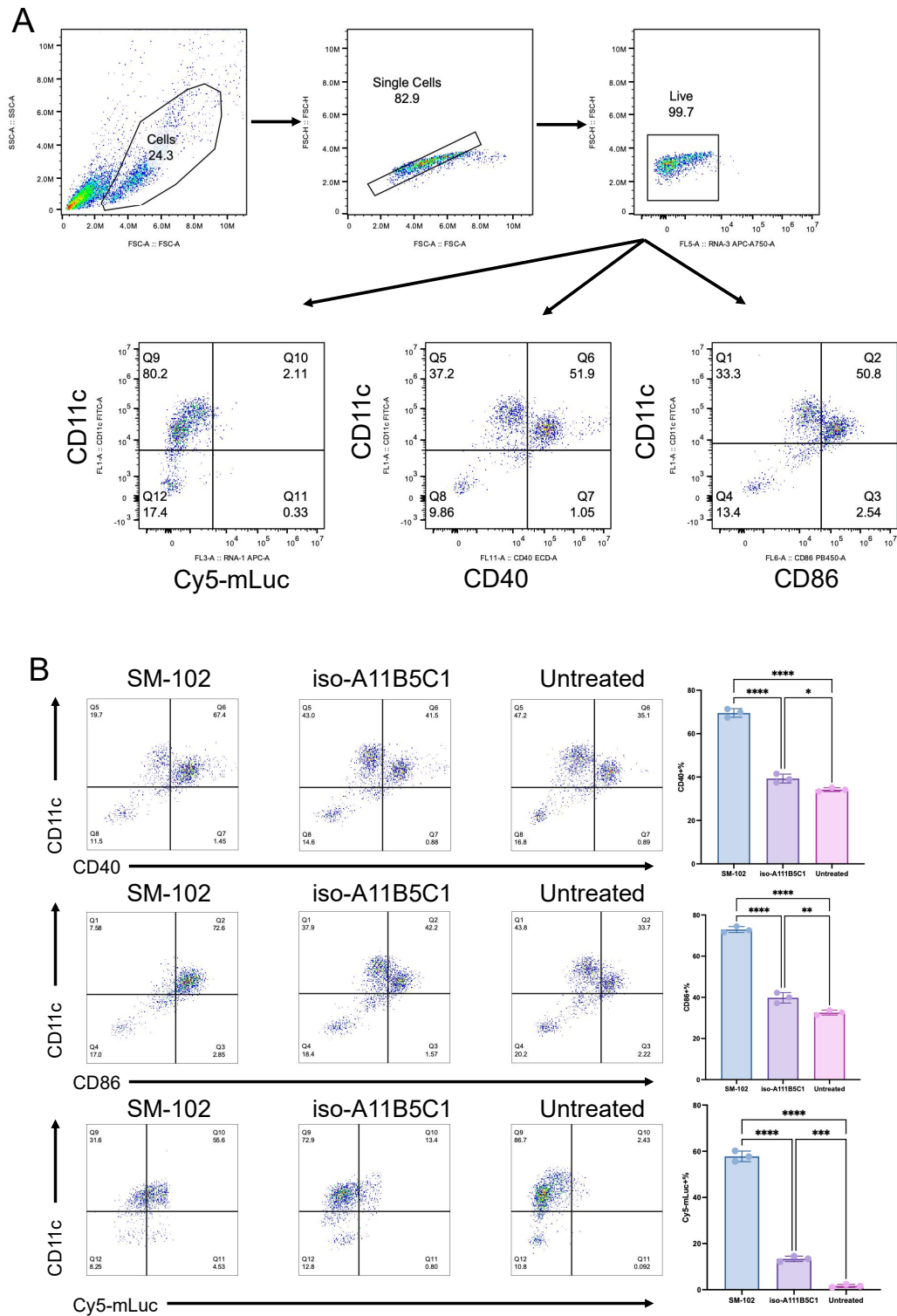


Fig. S9. (A) Gating information for BMDC staining. (B) Representative diagram and quantification of flow cytometry for analysis of Cy5-mLuc/LNP uptake by BMDC and BMDC subsequent activation.

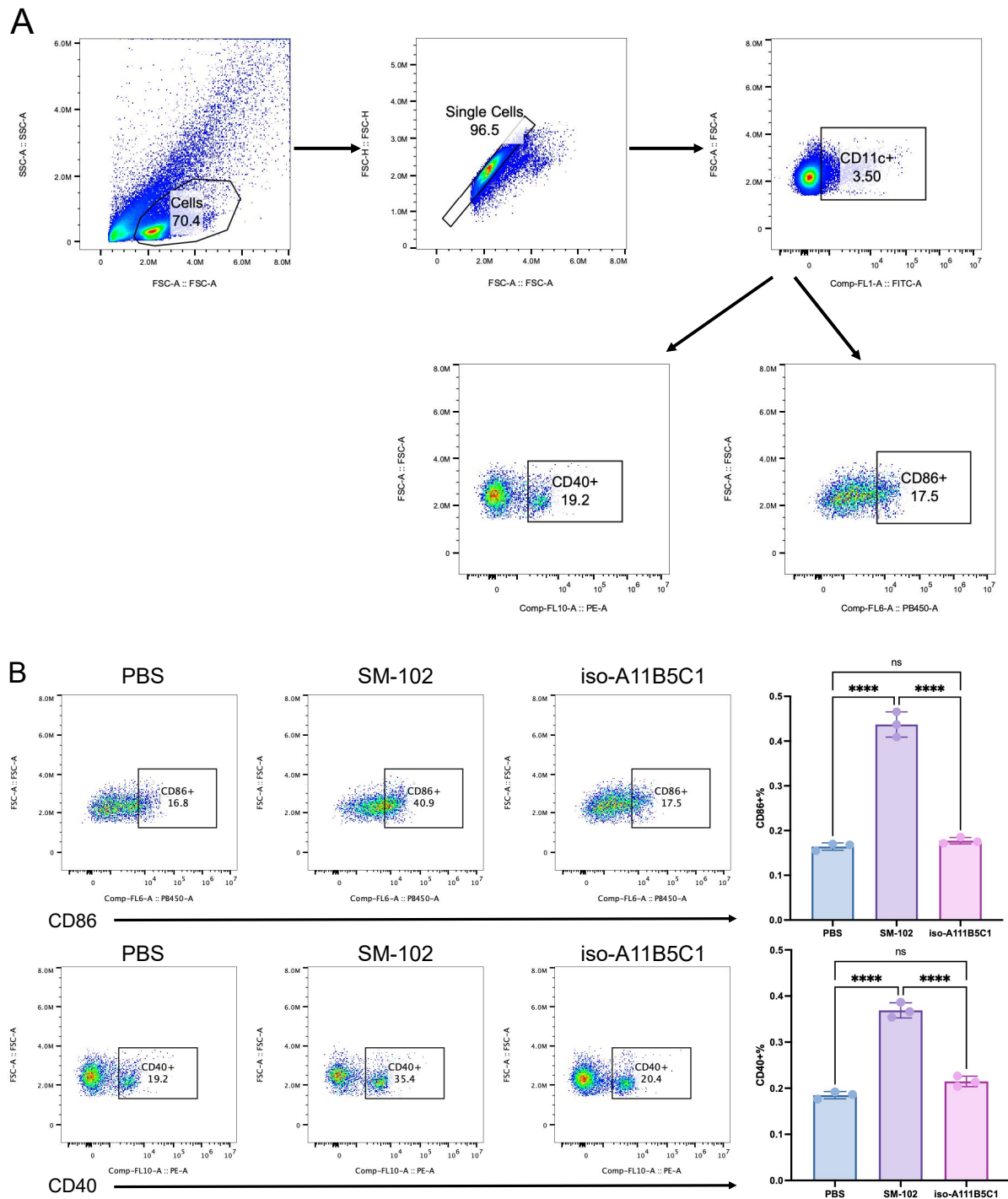


Fig. S10. (A) Gating information for lymph node staining. (B) Representative diagram and quantification of flow cytometry for analysis of the effect of LNP on the maturation of dendritic cells within inguinal lymph nodes.

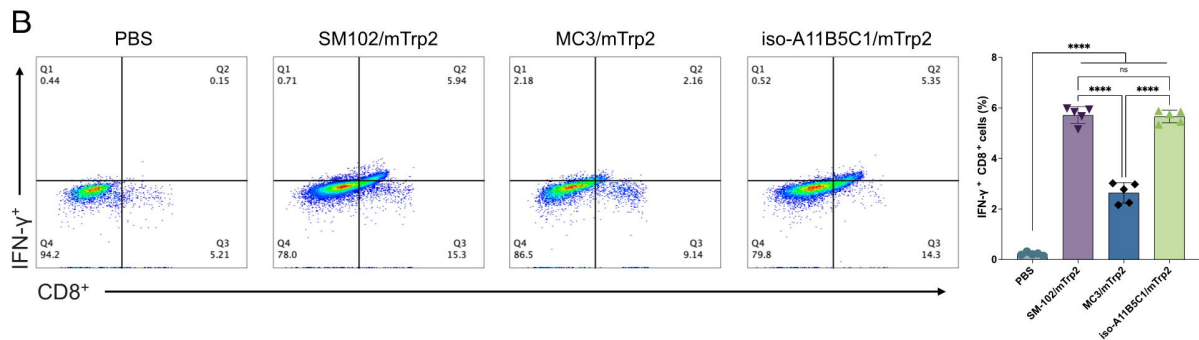
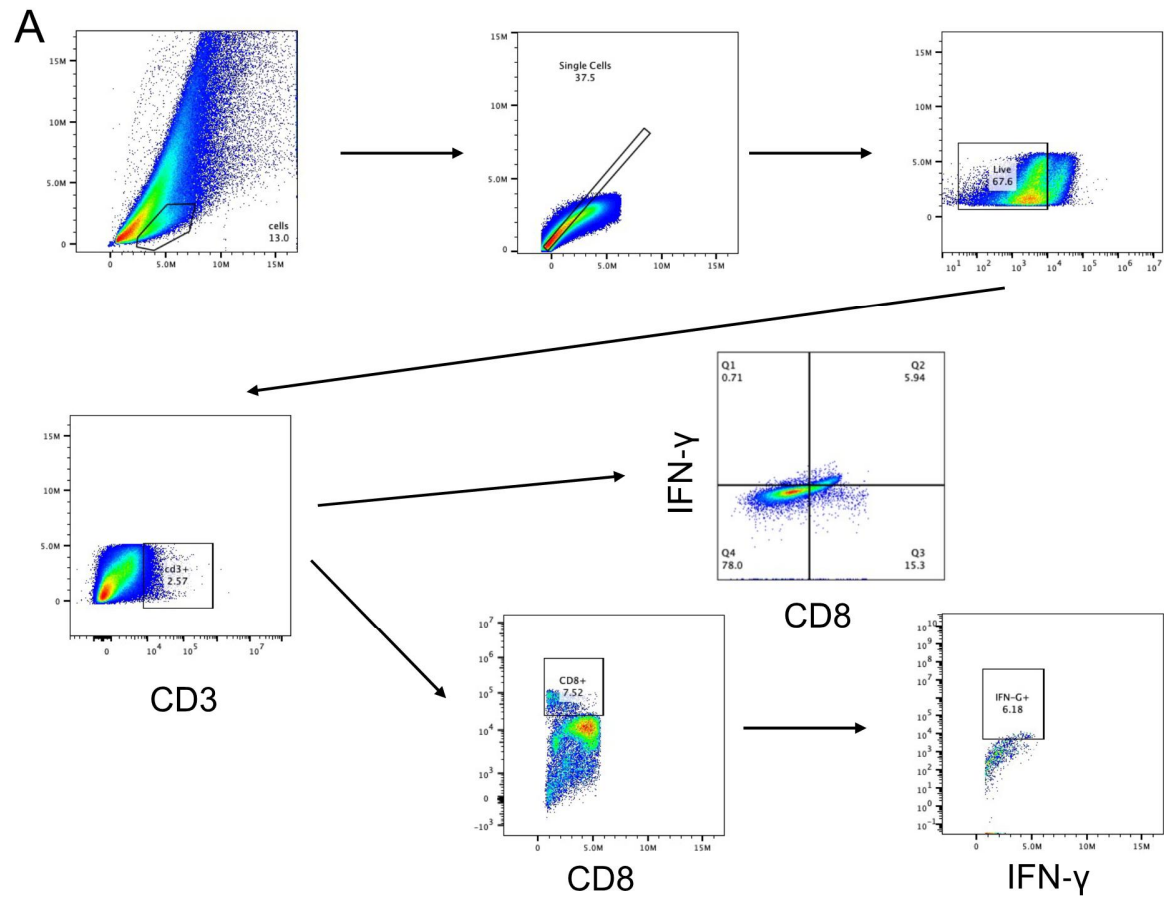


Fig. S11. (A) Gating information for B16-F10-Luc melanoma tumor cells. (B) Representative diagram and quantification of flow cytometry for analysis of the IFN- γ ⁺ CD8⁺ cells in the B16-F10-Luc tumor.

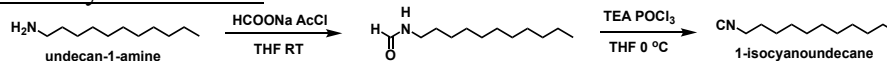
Table S3. Antibodies used for flow cytometry.

Target	Lable	Provider	Catalog
CD3	Brilliant Violet 605	BioLegend	100237
CD8	APC	BioLegend	100712
IFN- γ	Alexa Fluor® 488	BioLegend	505813
CD11c	FITC	BioLegend	117306
CD40	PE	Thermo Scientific	12-0401-81
CD86	Brilliant Violet 421™	BioLegend	105032
Zombie	NIR	BioLegend	423105
CD16/32	-	BioLegend	101320

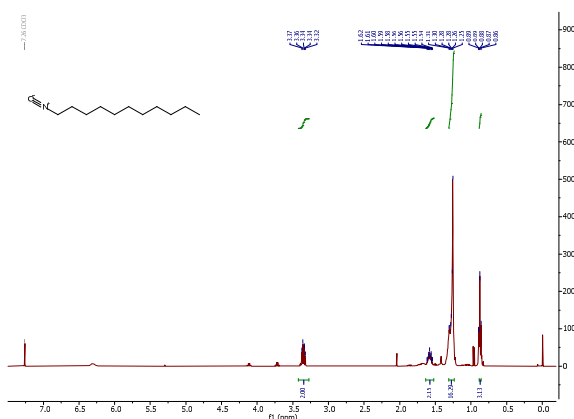
Fig. S12. NMR data for isocyanide tails

Isocyanide tail

Synthesis of 1-isocyanoundecane

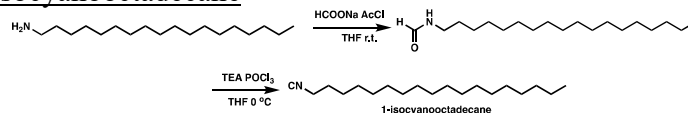


Iso-cyanoundecane was synthesized by the same procedure as the synthesis method above. ^1H NMR (400 MHz, CDCl_3) δ 3.42 – 3.27 (m, 2H), 1.63 – 1.52 (m, 2H), 1.31 – 1.24 (m, 16H), 0.88 – 0.86 (m, 3H).

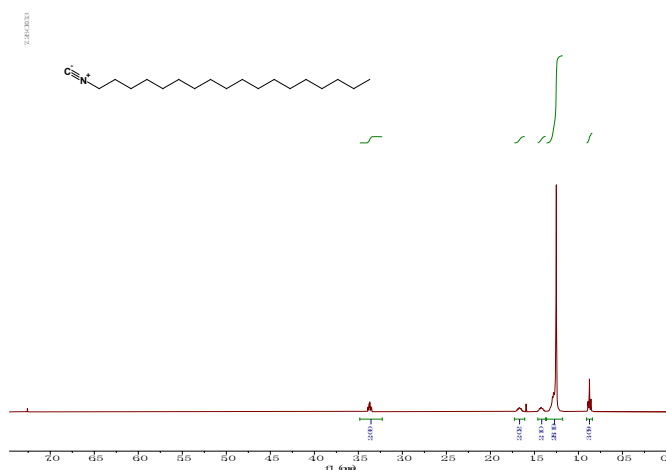


^1H -NMR of 1-isocyanoundecane

Synthesis of Tail 1-isocyanooctadecane



Follow the synthesis method above. ^1H NMR (400 MHz, CDCl_3) δ 3.48 – 3.23 (m, 2H), 1.73 – 1.61 (m, 2H), 1.46 – 1.37 (m, 2H), 1.25 (s, 28H), 0.91 – 0.84 (m, 3H).

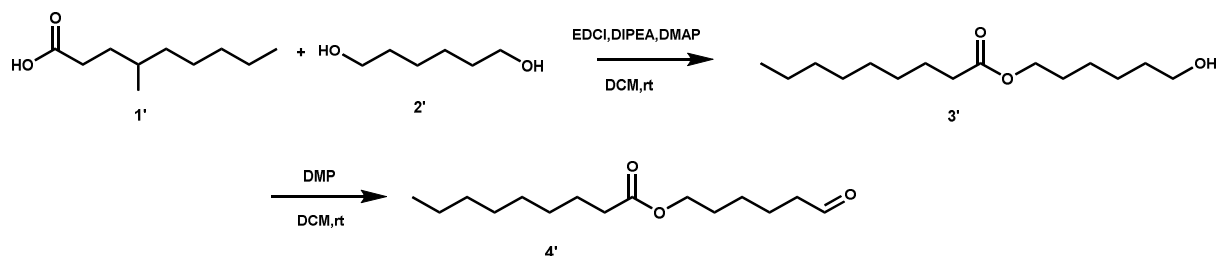


^1H -NMR of Iso-cyanooctadecane

Supporting Information Text

Chemical synthesis

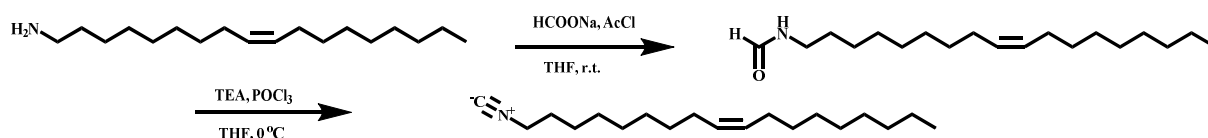
Synthesis of 6-oxohexyl 4-methylnonanoate. 4-methylnonanoic acid (3g, 17.7 mmol), *N*-(3-dimethylaminopropyl)- *N'*-ethylcarbodiimide hydrochloride (EDC HCl, 5.1g, 26.6 mmol), hexane-1,6-diol (10.5g, 88.5 mmol), 4-(dimethylamino) pyridine (DMAP, 1.1g, 8.9 mmol) and *N,N*-diisopropylethylamine (DIPEA, 6.2 ml, 35.4 mmol) were dissolved in dichloromethane (200 ml). The reaction was stirred at room temperature under nitrogen for 24 hours, then washed with a saturated aqueous sodium bicarbonate solution. The combined organic layers were washed with brine, dried over anhydrous Na₂SO₄, and then concentrated to obtain colorless oil crude product. The residue was purified by silica gel chromatography (0–50% ethyl acetate in hexanes) to give 6-hydroxyhexyl 4-methylnonanoate, **3'** (5.8g, 15.2 mmol, 86%) as a colorless oil. ¹H NMR (400 MHz, CDCl₃) δ 4.02 (dd, *J* = 7.2, 6.3 Hz, 2H), 3.59 (t, *J* = 6.6 Hz, 2H), 2.36 – 2.12 (m, 2H), 1.64 – 1.47 (m, 5H), 1.40 – 1.15 (m, 13H), 1.10 – 0.71 (m, 6H). 6-hydroxyhexyl 4-methylnonanoate, **3'** (5.8g, 15.2 mmol) was dissolved in dichloromethane 300 ml, followed by Dess-Martin Periodinane (9.6g, 22.8 mmol). The reaction was stirred under nitrogen at room temperature for 2 h. After confirmation of reaction completion by thin layer chromatography, sodium thiosulfate pentahydrate (50% w/v, 200 ml) was added to the reaction and left to stir for additional 15 mins. The combined organic layers were washed with brine, dried over anhydrous Na₂SO₄, and then concentrated to obtain colorless oil crude product. The residue was purified by silica gel chromatography (0–50% ethyl acetate in hexanes) to give 6-oxohexyl 4-methylnonanoate, **4'** (3.5g, 9.2 mmol, 61%) as a colorless oil. ¹H NMR (400 MHz, CDCl₃) δ 9.76 (t, *J* = 1.7 Hz, 1H), 4.05 (t, *J* = 6.6 Hz, 2H), 2.48 – 2.26 (m, 4H), 1.69 – 1.54 (m, 6H), 1.40 (tdd, *J* = 8.2, 6.5, 4.5 Hz, 4H), 1.29 – 1.21 (m, 8H), 0.86 (td, *J* = 6.6, 1.3 Hz, 6H).



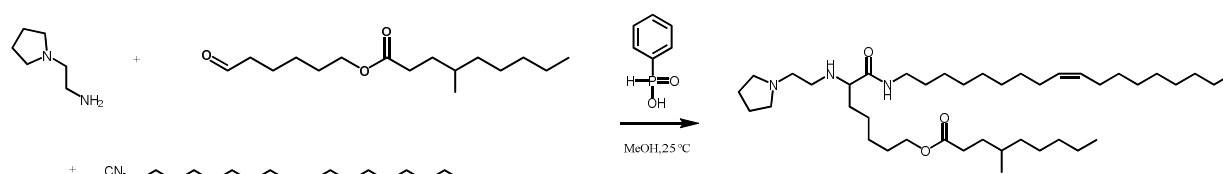
General synthetic route of 6-oxohexyl 4-methylnonanoate.

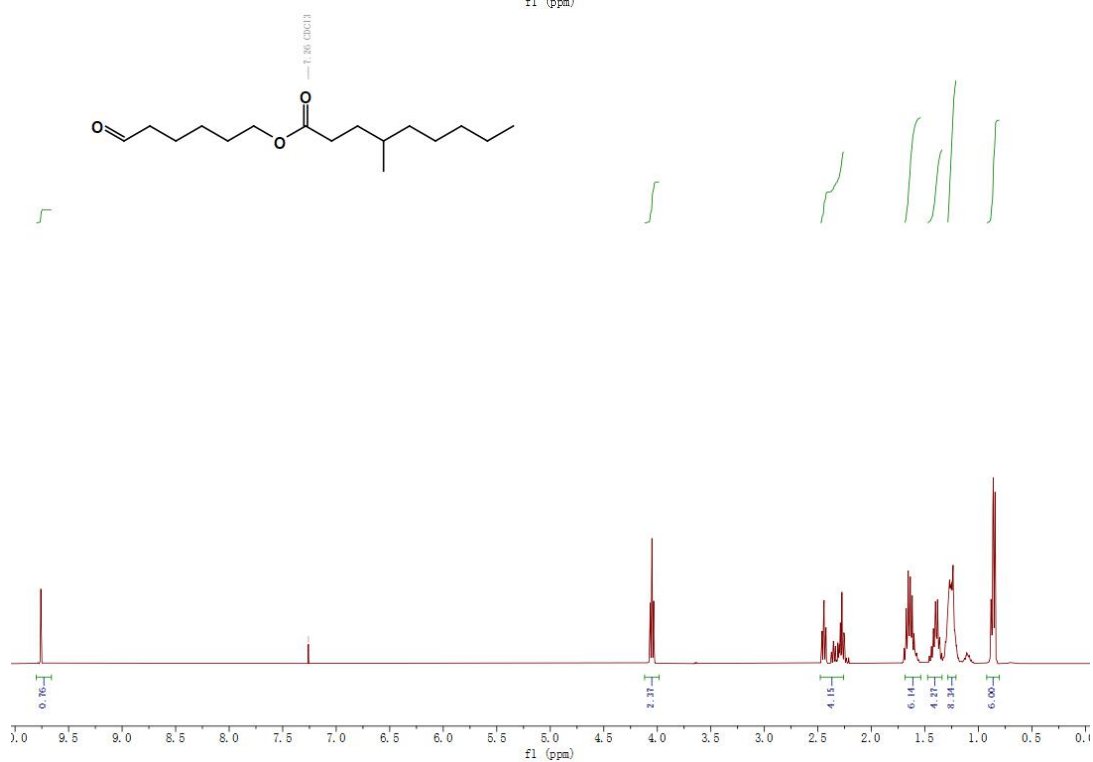
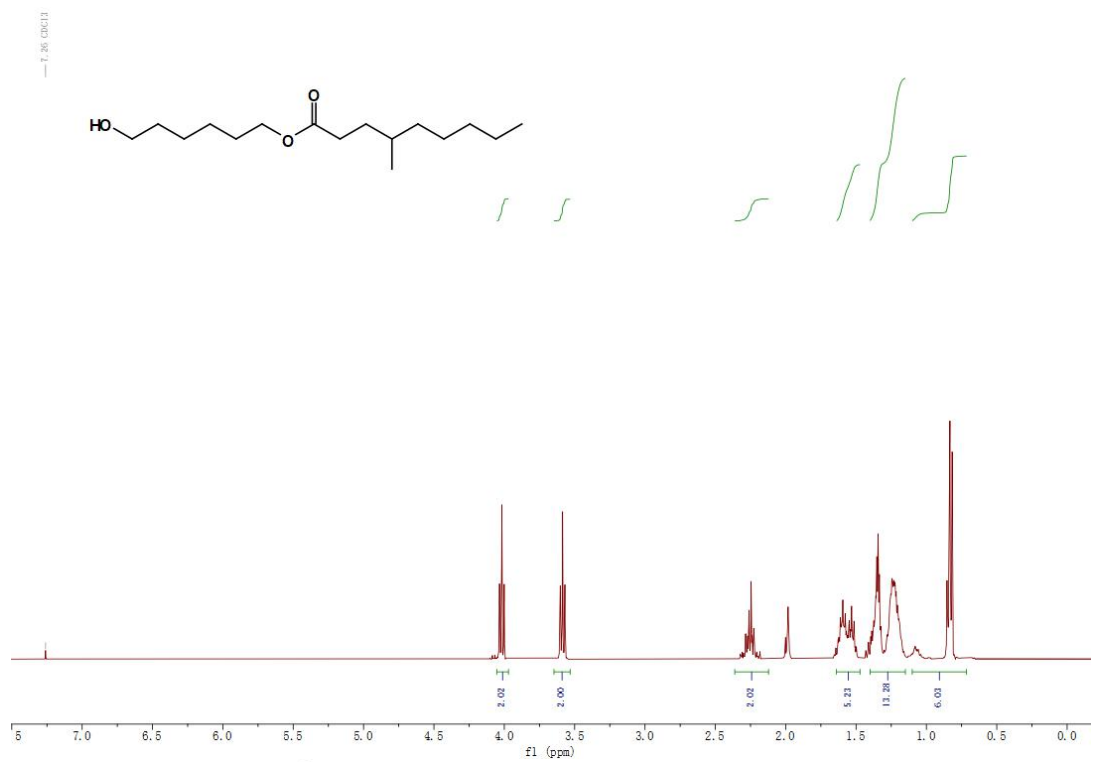
Synthesis of (Z)-1-isocyanooctadec-9-ene. Sodium formate (2.2511 g, 0.031 mol) was dissolved in THF (82.75 mL) into a round bottom flask and stir for 2 hours to form a suspension. Acetyl chloride (34.755 mL, 1.0 M in DCM, 0.031 mol) was added into the above-mentioned suspension by a constant pressure titration funnel and then stirred at room temperature for 12 hours to obtain solution 1. Oleyl-amine (8.8473 g, 0.031) was dissolved in THF (33.1ml) and added into solution 1, then stirred at room temperature for 4 hours after the dropwise addition to obtain solution 2. Solution 2 was diluted with 125 ml of deionized water, and the aqueous solution was extracted twice with ethyl acetate. The combined organic layers were washed with brine, dried over anhydrous Na₂SO₄, and then concentrated to obtain white solid crude product (Z)-N-(octadec-9-en-1-yl) formamide without any purification. Crude product (Z)-N-(octadec-9-en-1-yl) formamide (9.1581 g, 0.031 mol) in THF was added into the round-bottomed flask and stirred in an ice bath

at 0 °C for 10 min; TFA (0.031 mol) was added into the above solution and stirred in an ice bath for 15 min to obtain the solution 1. POCl₃ was dissolved in THF (50 mL) and added into solution 1 dropwise by a constant pressure titration funnel. After that, solution 1 was stirred for 1.5 hours at 0 °C. The reaction solution was washed with saturated sodium bicarbonate solution, then extracted by ether. The combined organic layers were washed with brine, dried over anhydrous Na₂SO₄, and then concentrated to obtain brown oil. The final product was purified by column chromatography using a BUCHI purification system with gradient elution (0–10% EA in Hexane) to give the corresponding products. (Z)-1-isocyanooctadec-9-ene yield 35%. ¹H NMR (400 MHz, CDCl₃) δ 3.48 – 3.23 (m, 2H), 1.73 – 1.61 (m, 2H), 1.46 – 1.37 (m, 2H), 1.25 (s, 28H), 0.91 – 0.84 (m, 3H).

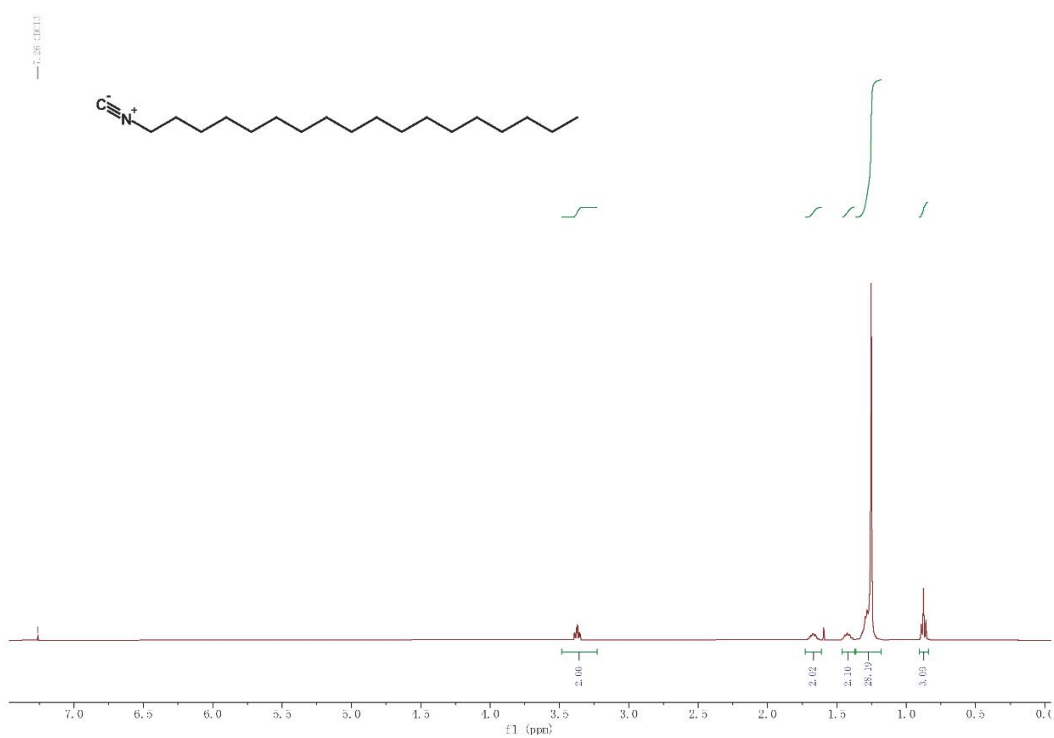
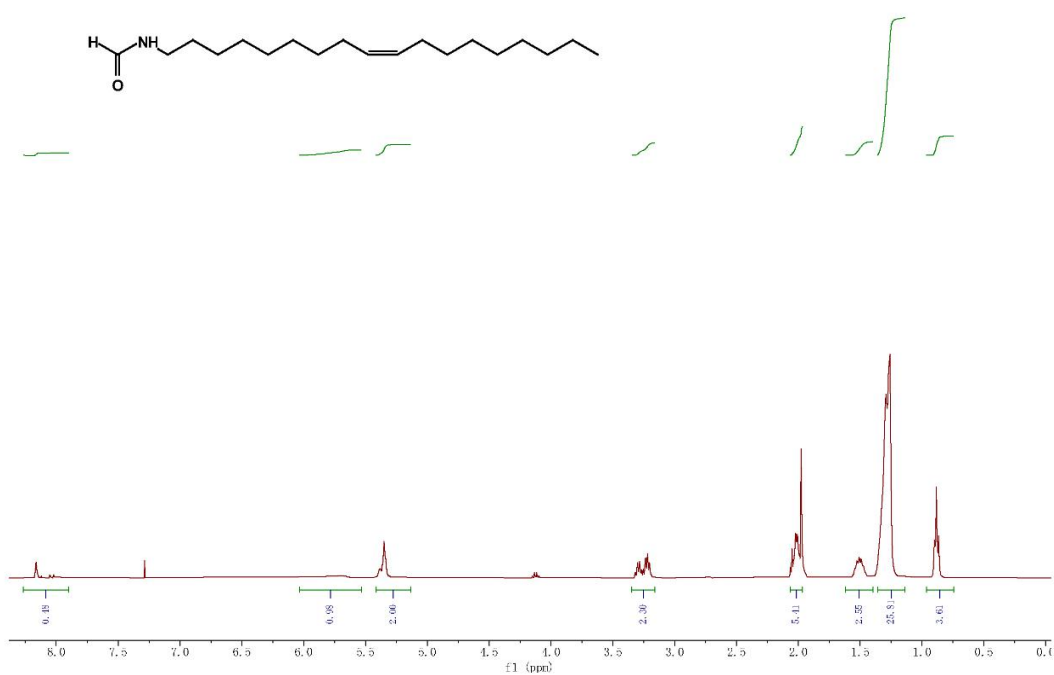


Synthesis of final product iso-A11B5C1. The synthesis of H3 through 3CR-Ugi reaction. A mixture of 2-(pyrrolidin-1-yl) ethan-1-amine (10 mg, 87.57 μmol), 6-oxohexyl 4-methylnonanoate (23.68 mg, 87.57 μmol), and (Z)-1-isocyanooctadec-9-ene (24.30 mg, 87.57 μmol) in anhydrous solvent MeOH (5 mL) with the catalyst phenylphosphinic acid (1.72 mg, 17.51 μmol) and was stirred in capped glass vials at room temperature overnight. The mixture was purified by flash column chromatography (0.1% ammonia hydroxide with 0-10% MeOH in DCM) to obtain compound H3 as pale-yellow oil (41 mg, 6.2 mmol, 70.71 %). MS calculated: m/z 661.61, observed (ESI ms) m/z 662.95. ¹H NMR (400 MHz, CDCl₃) δ 5.46 – 5.17 (m, 2H), 4.02 (t, *J* = 6.6 Hz, 2H), 3.22 (q, *J* = 6.8 Hz, 2H), 3.00 (s, 1H), 2.60 (d, *J* = 52.5 Hz, 6H), 2.35 – 2.16 (m, 2H), 1.98 (dq, *J* = 13.9, 6.4 Hz, 4H), 1.74 (s, 4H), 1.69 – 0.92 (m, 42H), 0.91 – 0.67 (m, 9H).

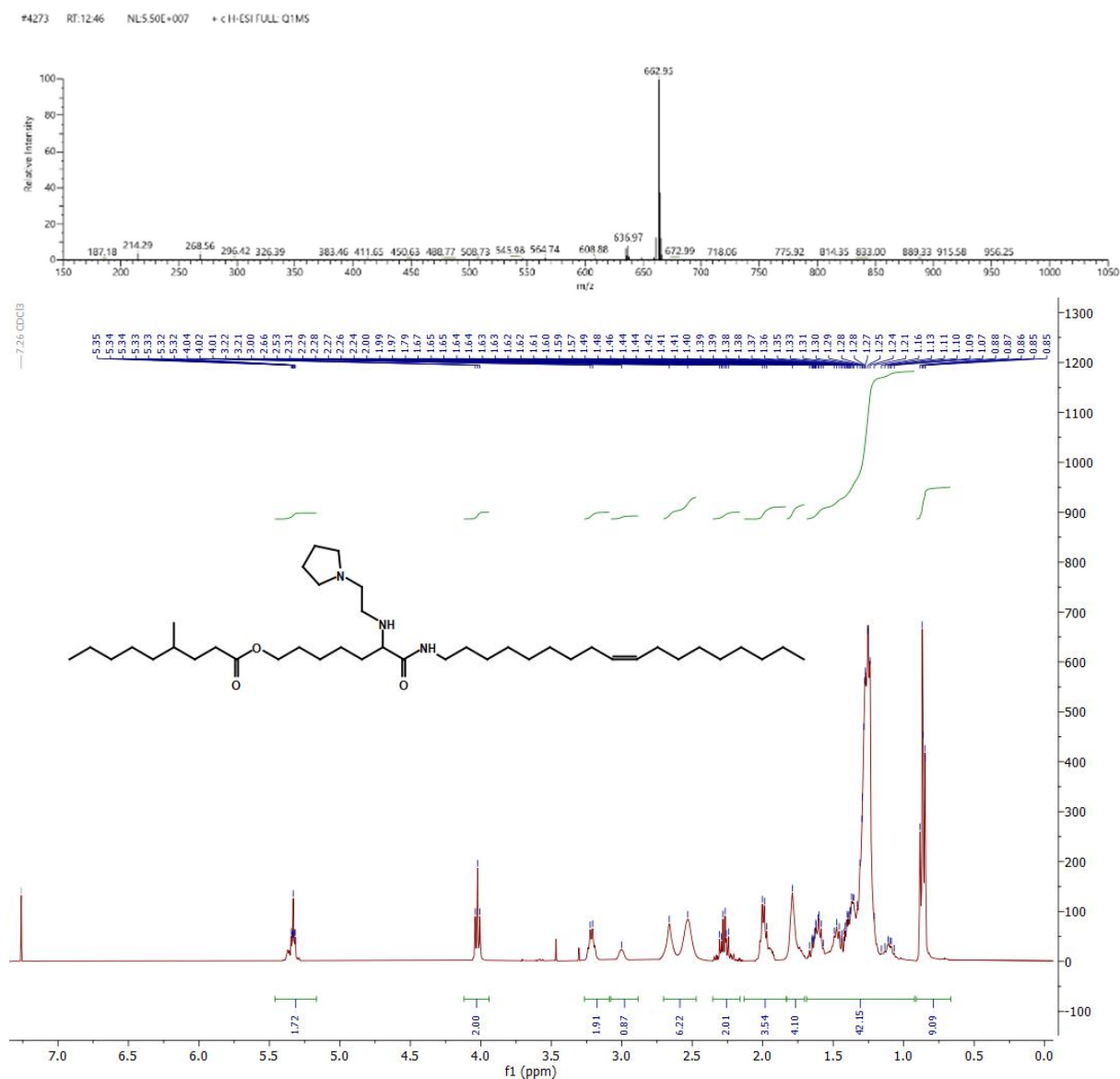




Spectrum (NMR and LC/MS) of 6-oxohexyl 4-methylnonanoate.



Spectrum (NMR and LC/MS) of (Z)-1-isocyanooctadec-9-ene.



Spectrum (NMR and LC/MS) of iso-A11B5C1.

Genbank sequence for full-length TRP2 plasmid

LOCUS TRP2_full_length 3802 bp DNA circular SYN 15-NOV-2023

DEFINITION synthetic circular DNA

FEATURES Location/Qualifiers

source 1..3802

/mol_type="other DNA"

/organism="synthetic DNA construct"

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/label=M13 rev

/note="common sequencing primer, one of multiple

similar variants"

rep_origin complement(245..833)

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of replication"

CDS complement(969..1784)

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/label=KanR

/note="confers resistance to kanamycin in bacteria

or G418 (Geneticin(R)) in eukaryotes"

/translation="MSHIQRETCSRPRLNSNMDADLYGYKWARDNVGQSGATIYRLYG

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CDS complement(969..1784)

/codon_start=1

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/note="confers resistance to kanamycin in bacteria

or G418 (Geneticin(R)) in eukaryotes"

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                                /note="vertebrate consensus sequence for strong
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